CHAPTER 4 · CONCEPT 3

The Heating Effect of Electric Current

Current meeting resistance turns electrical energy into heat — the principle behind heaters, kettles and irons.

Why a wire gets hot

When current flows through a conductor, the conductor offers some **opposition**, called **resistance**, to the flow. This resistance converts some electrical energy into **heat**. So the wire warms up — the **heating effect of electric current**.

> **Activity 4.5:** a **nichrome wire** feels warm when current passes (nichrome has higher resistance than copper of the same size, so it heats more).

What does the heat depend on?

- The **material** of the wire (high-resistance materials heat more).
- The **thickness** and **length** of the wire.
- The **amount of current** (more cells → more heat).
- The **time** for which current flows.

Everyday uses

Devices with a **heating element** (a coil/rod of high-resistance wire): room heater, electric stove, kettle, electric iron, immersion rod, hair dryer. An incandescent lamp glows because its filament is heated by current.

> **Two sides:** useful for heating, but overheating in wires/appliances can melt plastic parts or even cause fires. That's why circuits use correctly rated wires and safety devices.

Where students lose marks

> **Trap 1 — Nichrome, not copper, for heaters.** Nichrome has higher resistance, so it generates more heat for the same current. Copper is used for connecting wires (low resistance).

> **Trap 2 — "Good conductor" reasoning.** Nichrome is chosen because it gives more heat for a given current — not because it is the best conductor or because it is cheap.

> **Trap 3 — More current = more heat.** Two cells heat the wire more than one cell for the same time.

> **Safety:** never hold a hot nichrome wire; overheating can damage plugs/sockets or cause fire.

Model answers — write them like this

Example A. Why does a nichrome wire get hot when current flows?

Step 1: Nichrome offers high resistance to the current.

Step 2: This resistance converts electrical energy into heat.

∴ Resistance turns electrical energy into heat, so the wire warms up.

Example B. Why is nichrome, not copper, used in an electric heater's element?

Step 1: For the same size, nichrome has higher resistance than copper.

Step 2: Higher resistance → more heat for the same current.

∴ Nichrome produces more heat, so it is used for heating elements.

Ready to practise? → Open the worksheet